

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

18/20

Code of Conduct:

I V. Sanjay (192312486) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Sanjay
Signature of the student:



Annexure A



Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

Smart Campus Subnetting:

Given IP block: 192.168.100.0/24.

Department	Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
School of Computing	/25	192.168.100.0	192.168.100.1 - 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 - 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.193 - 192.168.100.222	192.168.100.223	30

Remaining unused IP range:

The original /24 block ends at 192.168.100.255.

⇒ Allocated up to 192.168.100.223.

⇒ Unused range: 192.168.100.224 - 192.168.100.255.

⇒ This includes 32 IP addresses

⇒ If treated as a subnet, it is 192.168.100.224/27.

⇒ Usable hosts in this unused/27:

192.168.100.225 - 192.168.100.254 (30 usable hosts).

② International Software Company

Subnetting:

IP block: 10.10.0.0/16

Dept	Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
Software Development	/24	10.10.0.0	10.10.0.1 - 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 - 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 - 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 - 10.10.1.222	10.10.1.223	30

Remaining unused IP range:

The /16 block is:

10.10.0.0 - 10.10.255.255

Allocated range ends:

10.10.1.223

The remaining unused range is:

10.10.1.224 - 10.10.255.255

Total addresses in /16 = 65536

Allocated addresses:

⇒ /24 = 256

⇒ /25 = 128

⇒ /26 = 64

⇒ /27 = 32

⇒ Total = 480

Remaining addresses = 65536 - 480 = 65056

Smart Hospital Network.

IPv6 address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation.

Remove leading zeros in each group and replace consecutive zero groups with

Ans: 2001:db8:abcd::/64.

2. Expanded full hexadecimal representation.

2001:0db8:abcd::/64.

Expanded:

2001:0db8:abcd:0000:0000:0000:0000:0000/64.

3. Network prefix:

The prefix length is /64. So the first 64 bits represent the network.

2001:0db8:abcd:0000::/64.

2001:0db8:abcd:0000

4. Total possible host addresses.

IPv6 has 128 bits.

Network bits = 64.

Host bits.

Total host addresses.

$$2^{64} = 18,446,744,073,409,851,616.$$

④ ISP Routing Table:

192.168.1.0/24.

192.168.2.0/24.

192.168.3.0/24.

Destination IP:

192.168.2.45

1. Destination subnet.

the destination 192.168.2.45 belongs to:
192.168.2.0/24.

192.168.2.0 - 192.168.2.255.

2. Matching routing table entry.

192.168.2.0/24

this is the matching route.

3. Next hop network / interface:

192.168.2.0/24 network.

default route

0.0.0.0/0.

The actual next-hop address for the default route is not specified in the question.

Administration and Finance LANs.

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Administration Gateway: 192.168.10.1

Finance Gateway: 192.168.10.65.

Each /26 subnet has:

$$2^{(32-26)} = 64 \text{ total addresses}$$

Usable hosts:

$$64 - 2 = 62.$$

Suitable IP addresses for three computers

Computer 1 $\rightarrow$ 192.168.10.2

" 2 $\rightarrow$ 192.168.10.3

" 3 $\rightarrow$ 192.168.10.4.

Finance LAN.

Computer 1 $\rightarrow$ 192.168.10.66.

2 $\rightarrow$ 192.168.10.67

3 $\rightarrow$ 192.168.10.68.

255.255.255.192.

Default Gateway:

HAN

Default Gateway.

Administration

192.168.10.1

Finance

192.168.10.65.

Final Summary:

→ Administration: 192.168.10.0/24 → Hosts

→ .1 - 62 → Gateway .1 → Broadcast .63.

→ Finance : 192.168.10.64/28 → Hosts.

.65 - 126 → Gateway .65 → Broadcast .127.

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